



MATHEMATICS EXTENSION 1

PRELIMINARY EXAMINATION

2004

Time Allowed – 1.5 hours
(Plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES

- Attempt all Questions.
- Start each Question on a new page.
- Board approved calculators may be used.
- All necessary working should be shown.

OUTCOMES ASSESSED

PE1, PE2, PE3, PE4, PE5, PE6.

QUESTION 1 (12 Marks)

Marks

- a). Solve for x :

$$\frac{3x-1}{x-2} \geq 1$$
3
- b). Find the acute angle between the lines $3x + y = 4$ and $2x - 3y = 1$.
 (Write your answer to the nearest minute). 2
- c). Find the exact value of $\sin 75^\circ$ 2
- d). The equation $x^2 + px + 24 = 0$ has one root which is one and a half times the other. Show that there are two possible values for p and find the roots of each of these values. 3
- e). Write down the equation of the locus of a parabola with vertex at (2,3) and focus (2,1). 2

QUESTION 2 (12 Marks)

Use a new page

- a). Find $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$ 2
- b). If $f(x) = \frac{1}{\sqrt{x - x^2}}$ find the domain. 1
- c). Find the general solution to the equation $2\cos(2x - 60^\circ) = \sqrt{3}$ 2
- d). The point $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$
 (i). Show that the equation of the tangent at P is $y = px - ap^2$ 2
 (ii). If $Q(2aq, aq^2)$ also lies on the parabola, and PQ is a focal chord, show that the tangents at P and Q intersect on the directrix at right angles. 3
- e). How many different arrangements of the letters of the word *SERIES* are possible if
 (i). the two letters S are to be together.
 (ii). no two vowels are together. 2

QUESTION 3 (12 Marks) Use a new page **Marks**

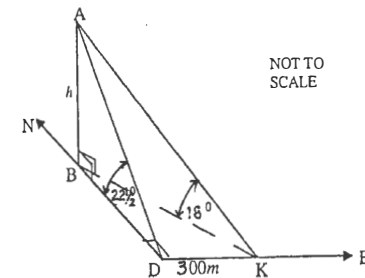
- a).
 (i). In how many ways can 6 men and 5 women sit around a circular table without restrictions? **2**
 (ii). How many ways are there if 2 particular women wish to sit next to a particular man? **2**
- b). For the curve $y = \frac{x^2 - x + 1}{x - 1}$
 (i). Find any stationary points and determine their nature.
 (ii). Write down the equations of any asymptotes.
 (iii). Write down any intercepts made with the co-ordinate axes.
 (iv). Sketch the curve. **8**

QUESTION 4 (12 Marks) Use a separate page **Marks**

- a). (i). Write the expansion for $\cos(\alpha + \theta)$ **1**
 (ii). Hence, or otherwise, prove that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ **2**
 (iii). Solve $8 \cos^3 \theta - 6 \cos \theta - \sqrt{3} = 0$ for $0^\circ \leq \theta \leq 360^\circ$ **2**
- b). Prove $\frac{\tan x - \tan y}{\tan x + \tan y} = \frac{\sin(x - y)}{\sin(x + y)}$ **3**
- c). If P (6, 2) divides the interval AB externally in the ratio 2:7, find the co-ordinates of B, if A has the co-ordinates (4, 0). **2**
- d). Find the co-ordinates of Q, given that the chord of contact of the tangents from Q to the parabola $x^2 = 14y$ is $5x - 14y - 21 = 0$ **2**

QUESTION 5 (12 Marks) Use a new page **Marks**

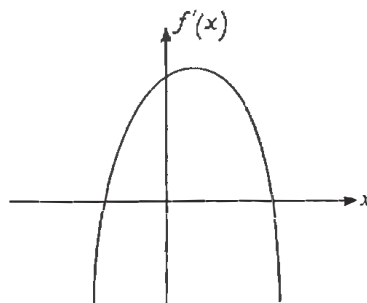
- a). Doris is standing at D and observes the angle of elevation of the tip of a flagpole A, on top of a building to be $22\frac{1}{2}^\circ$. Her friend Katie, who is standing at K, 300 metres due east of Doris, finds the angle of elevation of the tip of the flagpole to be 18° . The building is due north of Doris and B is the base of the building. The points B, D and K are on level ground.



- (i) Show that the height (h) of the flagpole above the ground is given by $h = 300 (\cot^2 18^\circ - \cot^2 22\frac{1}{2}^\circ)^{1/2}$ **3**
 (ii) Find the value of h , correct to 3 significant figures **1**
- b) (i) Express $\sqrt{3} \cos \theta + \sin \theta$ in the form $A \cos(\theta - \alpha)$ where $A > 0$ **2**
 (ii) Hence, solve the equation $\sqrt{3} \cos \theta + \sin \theta = -\sqrt{3}$ for $0^\circ \leq \theta \leq 360^\circ$ **2**
- c) (i) Given that $\frac{d^2 y}{dx^2} = x - 1$, and the tangent to the curve $y = f(x)$ at the point (0, -1) is inclined at 45° to the positive x -axis. Find the equation of the curve. **2**
 (ii) Find $\frac{d^2 y}{dx^2}$ for the function $y = \sqrt{1 - 2x}$ **2**

QUESTION 6 (12 Marks) Use a new page **Marks**

a). Given the gradient function in the diagram below:- **2**



Draw a set of axes and sketch a function to represent $f(x)$.

b) (i) If $f(x) = \frac{\sqrt{x}+1}{\sqrt{x}-1}$, show that $f'(x) = \frac{-1}{\sqrt{x}(\sqrt{x}-1)^2}$ **2**

(ii) Show that the equation of the tangent at (4,3) on the curve

$f(x) = \frac{\sqrt{x}+1}{\sqrt{x}-1}$ is given by $x + 2y - 10 = 0$ **2**

c) The running cost (cost of fuel) for a certain ship is \$4 per hour when the ship is not moving, and this cost increases by an amount that is proportional to the cube of its speed, V km/h. If the running cost per hour is \$7.25 when the speed is 15 km/h, obtain a formula for the running cost per hour at speed V , and calculate the value of V for which the total running cost for a journey of 500km is a minimum. **4**

d) Find the Cartesian equation of the curve whose parametric equations are $x = 1 + 2^t$ and $y = 2^t$. **2**

Question 1

a) Solve $\frac{3x-1}{x-2} \geq 1$ $x \neq 2$ ①

For $x-2 > 0$
 $x > 2$

$3x-1 \geq x-2$

$2x \geq -1$

$x \geq -\frac{1}{2}$

Simultaneously: $x > 2$



For $x-2 < 0$
 $x < 2$

$3x-1 \leq x-2$

$2x \leq -1$

$x \leq -\frac{1}{2}$

Simultaneously: $x \leq -\frac{1}{2}$



Solution is $x \leq -\frac{1}{2}$ or $x > 2$ ③

b) Find the acute angle between $3x+y=4$ & $2x-3y=1$.

$l_1: y = 4-3x$

$l_2: y = \frac{2x-1}{3}$

$\therefore m_1 = -3$

$\therefore m_2 = \frac{2}{3}$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $= \left| \frac{-3 - \frac{2}{3}}{1 + (-3)(\frac{2}{3})} \right|$

$= \frac{11}{3}$

$\therefore \theta = 74^\circ 45'$

Thus, the acute angle is $74^\circ 45'$ ②

c) Find $\sin 75^\circ$.

$\sin 75^\circ = \sin (30^\circ + 45^\circ)$

$= \sin 30^\circ \cos 45^\circ + \sin 45^\circ \cos 30^\circ$

$= \frac{1}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2}$

$\therefore \sin 75^\circ = \frac{1+\sqrt{3}}{2\sqrt{2}}$ ①

d) Let $x^2 + px + 24 = 0$ have 2 roots α, β , where $\frac{3}{2}\alpha = \beta$

then sum of roots: $\frac{3}{2}\alpha + \alpha = -\frac{p}{1}$ ①

product of roots: $\frac{3}{2}\alpha^2 = 24$
 $\frac{3}{2}\left(-\frac{2p}{5}\right)^2 = 48$

$p^2 = 100$

$\therefore p = \pm 10$ ③

The 2 resulting quadratics are

$x^2 - 10x + 24 = 0$ where the roots are $x = 4, 6$

$x^2 + 10x + 24 = 0$ where the roots are $x = -4, -6$.

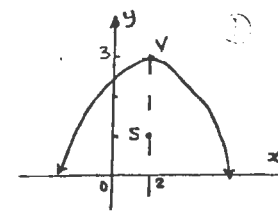
e) Focus $(2, 1)$, Vertex $(2, 3)$

$\therefore a = 2$

Equation has the form:

$(x-h)^2 = -4a(y-k)$

$\therefore (x-2)^2 = -8(y-3)$ ②

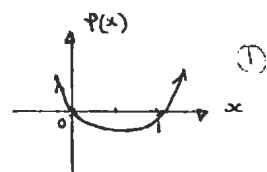


Question 2

a) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2} = \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)}{x-2}$ for $x \neq 2$ ①
 $= \lim_{x \rightarrow 2} x^2 + 2x + 4$
 $= 2^2 + 2 \times 2 + 4$
 $= 12$ 1/2

b) Given $f(x) = \frac{1}{\sqrt{x-x^2}}$

then $x - x^2 > 0$
 ie $x^2 - x < 0$
 $x(x-1) < 0$
 $\therefore 0 < x < 1$
 Domain = $\{x: 0 < x < 1\}$ 1



c) $2 \cos(2x - 60^\circ) = \sqrt{3}$ $\cos 30^\circ = \frac{\sqrt{3}}{2}$ ②
 $2x - 60^\circ = n \cdot 360^\circ \pm 30^\circ$
 $2x = n \cdot 360^\circ + 90^\circ, n \cdot 360^\circ + 30^\circ$
 $\therefore x = n \cdot 180^\circ \pm 15^\circ + 30^\circ$ 1/2

(i) Show the eqn. of the tangent at P is $y = px - ap^2$.

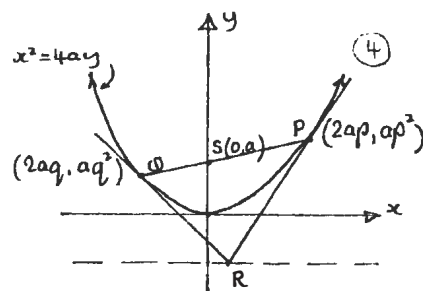
$y = \frac{x^2}{4a}$

$y' = \frac{x}{2a}$ at $x = 2ap$

$m = p$

$y - ap^2 = p(x - 2ap)$

$y = px - ap^2$ (as required)



b) (iii) Intercepts: if $y = 0$ then $x^2 - x + 1 = 0$
 $x - 1$

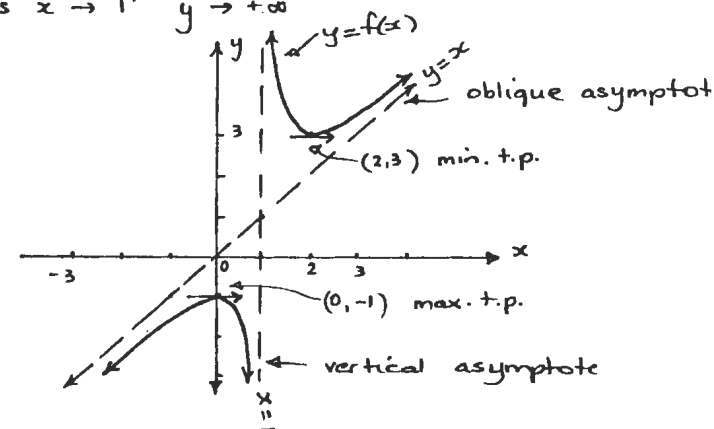
has no solutions, thus there are no x intercepts.
 if $x = 0 \therefore y = -1$.

(iv) Sketch

Check the curve near the asymptote $x = 1$

as $x \rightarrow 1^-$ $y \rightarrow -\infty$

as $x \rightarrow 1^+$ $y \rightarrow +\infty$



Question 3a

- (i) The first person sits anywhere, and then treating the circle now as a line, the remaining 10 people sit in any spot. That is $10!$ possibilities.
- (ii) Consider the woman-man-woman group as a single 'unit' in the circle (two possible arrangements), leaving 5 men and 3 women i.e. 8 people.
Using a similar argument to (i), in total there are $8! \times 2$ possibilities (the 2 comes from arrangement within the group)

Question 3b

- (i) *Stationary points:*

$$y = \frac{x^2 - x}{x - 1} + \frac{1}{x - 1} = x + \frac{1}{x - 1}$$

$$\frac{dy}{dx} = 1 - \frac{1}{(x - 1)^2} = \frac{(x - 1)^2 - 1}{(x - 1)^2} = \frac{x(x - 2)}{(x - 1)^2}$$

$$\frac{d^2y}{dx^2} = \frac{2}{(x - 1)^3}$$

Therefore, stationary points occur at $x = 0$ and $x = 2$.

The second derivative at $x = 0$ is negative, so $x = 0$ is a **local maxima**,

The second derivative at $x = 2$ so $x = 2$ is a **local minima**.

- (ii) *Asymptotes:*

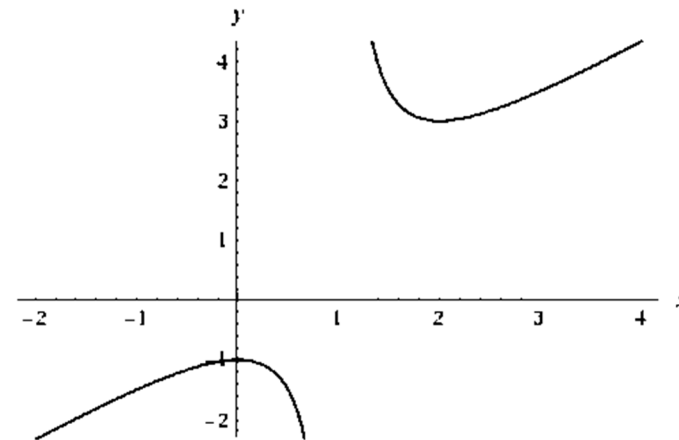
The equation of the asymptote is $x = 1$

- (iii) *Intercepts:*

There is no x -intercept as the discriminant of $x^2 - x - 1$ is negative which implies that $x^2 - x - 1 = 0$ has no solution.

The y -intercept is $y = -1$

- (iv) *Graph of function:*



Question 4

a) (i) $\cos(\alpha + \theta) = \cos \alpha \cos \theta - \sin \alpha \sin \theta$.

(ii) Prove $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$

Proof: L.H.S. = $\cos 3\theta$
 $= \cos(2\theta + \theta)$
 $= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$
 $= (\cos^2 \theta - \sin^2 \theta) \cos \theta - 2 \sin \theta \cos \theta \sin \theta$
 $= \cos^3 \theta - \sin^2 \theta \cos \theta - 2 \sin^2 \theta \cos \theta$
 $= \cos^3 \theta - 3 \sin^2 \theta \cos \theta$
 $= \cos^3 \theta - 3(1 - \cos^2 \theta) \cos \theta$
 $= 4 \cos^3 \theta - 3 \cos \theta$.

$\therefore$ L.H.S. = R.H.S.

(iii) Solve $8 \cos^3 \theta - 6 \cos \theta - \sqrt{3} = 0$ $0^\circ \leq \theta \leq 360^\circ$.

divide b.s. by 2 giving
 $4 \cos^3 \theta - 3 \cos \theta = \frac{\sqrt{3}}{2}$

Now, from above

$4 \cos^3 \theta - 3 \cos \theta = \cos 3\theta$

$\therefore \cos 3\theta = \frac{\sqrt{3}}{2}$

$3\theta = 30^\circ, 330^\circ, 390^\circ, 690^\circ, 750^\circ, 1050^\circ$

$\therefore \theta = 10^\circ, 110^\circ, 130^\circ, 230^\circ, 250^\circ, 350^\circ$.

b) Prove $\tan x - \tan y = \frac{\sin(x-y)}{\cos(x+y)}$

L.H.S. = $\frac{\sin x}{\cos x} - \frac{\sin y}{\cos y}$
 $= \frac{\sin x \cos y - \sin y \cos x}{\cos x \cos y}$
 $= \frac{\sin x \cos y - \sin y \cos x}{\cos x \cos y}$
 $= \frac{\sin x \cos y - \sin y \cos x}{\sin x \cos y + \sin y \cos x}$
 $= \sin(x-y)$

c) P(6,2) divides AB externally in the ratio 2:7

If A(4,0) find B(x₂, y₂)

$6 = \frac{2(x_2) - (7)(4)}{2-7} \quad 2 = \frac{2(y_2) - (7)(0)}{2-7}$

$\therefore x_2 = -1$

$y_2 = -5$

Thus, the coordinates of B are (-1, -5)

d) Find the coordinates of Q, given that the chord of contact of the tangents from Q to $x^2 = 14y$ is $5x - 14y - 21 = 0$.

The chord of contact has the form:

$xx_0 = 2a(y + y_0)$ where $(x_0, y_0) = Q$

$\therefore xx_0 = 7y + 7y_0$

$a = \frac{7}{2}$

i.e. $xx_0 - 7y - 7y_0 = 0$

given $5x - 14y - 21 = 0$

The ratio of the coefficients are equal

$\therefore \frac{x_0}{5} = \frac{-7}{-14} \quad \frac{-7y_0}{-21} = \frac{-7}{-14}$

$\therefore x_0 = 5/2$

$y_0 = 3/2$

Thus, Q has coordinates (5/2, 3/2)

Question 3

a) (i) show $h = 300(\cot^2 18^\circ - \cot^2 22\frac{1}{2}^\circ)^{-\frac{1}{2}}$. (3)

From $\triangle BAD$: $\cot 22\frac{1}{2}^\circ = \frac{BD}{h}$

$\therefore BD = h \cot 22\frac{1}{2}^\circ$

From $\triangle BAK$: $\cot 18^\circ = \frac{BK}{h}$

$\therefore BK = h \cot 18^\circ$

Using Pythagoras' Theorem:-

$$h^2 \cot^2 18^\circ - h^2 \cot^2 22\frac{1}{2}^\circ = 300^2$$

$$\therefore h = 300(\cot^2 18^\circ - \cot^2 22\frac{1}{2}^\circ)^{-\frac{1}{2}}$$

(ii) $h = 157.16268\dots$
 ≈ 157 metres

b) (i) Express $\sqrt{3} \cos \theta + \sin \theta$ in the form $A \cos(\theta - \alpha)$.

$$A \cos(\theta - \alpha) = A(\cos \theta \cos \alpha + \sin \theta \sin \alpha)$$

$$= \sqrt{3} \cos \theta + \sin \theta$$

Equating like parts,

$$A \cos \alpha = \sqrt{3} \quad A \sin \alpha = 1 \quad \tan \alpha = \frac{1}{\sqrt{3}}$$

$$A^2(\cos^2 \alpha + \sin^2 \alpha) = \sqrt{3}^2 + 1^2 \quad \therefore \alpha = 30^\circ$$

$$A^2 = 4$$

$$\therefore A = 2 \text{ (take +ve root only!)}$$

$$\therefore \cos \alpha = \frac{\sqrt{3}}{2} \quad \& \quad \sin \alpha = \frac{1}{2}$$

$$\therefore \sqrt{3} \cos \theta + \sin \theta = 2 \cos(\theta - 30^\circ)$$

(ii) Solve $\sqrt{3} \cos \theta + \sin \theta = -\sqrt{3} \quad 0^\circ \leq \theta \leq 360^\circ$

From (i) $2 \cos(\theta - 30^\circ) = -\sqrt{3}$

$$\therefore \cos(\theta - 30^\circ) = -\frac{\sqrt{3}}{2}$$

$$\theta - 30^\circ = 150^\circ, 210^\circ$$

$$\therefore \theta = 180^\circ, 240^\circ$$

Diagram showing a unit circle with angles 180-30 and 180+30, and a value of -sqrt(3)/2 for cos 30.

c) (i) $\frac{d^2y}{dx^2} = x - 1$ ①

$$\frac{dy}{dx} = \frac{x^2}{2} - x + C \quad \text{when } x=0 \quad \frac{dy}{dx} = -1$$

$$-1 = C$$

$$\therefore \frac{dy}{dx} = \frac{x^2}{2} - x - 1$$

$$y = \frac{x^3}{6} - \frac{x^2}{2} - x + K \quad \text{when } x=0, y=-1$$

$$-1 = K$$

$$\therefore y = \frac{x^3}{6} - \frac{x^2}{2} - x - 1$$

(ii) $y = \sqrt{1-2x}$
 $= (1-2x)^{\frac{1}{2}}$
 $\frac{dy}{dx} = \frac{1}{2}(1-2x)^{-\frac{1}{2}} \cdot -2$

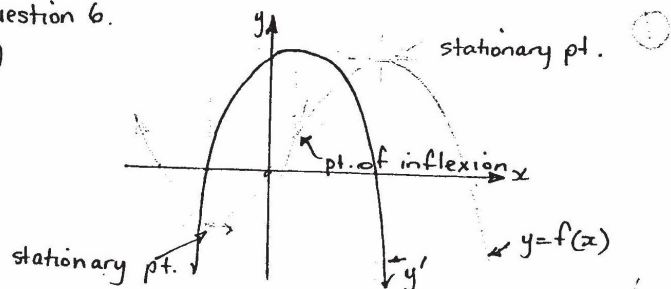
$$= \frac{-1}{\sqrt{1-2x}}$$

$$\frac{d^2y}{dx^2} = -1 \cdot -\frac{1}{2}(1-2x)^{-\frac{3}{2}} \cdot -2$$

$$= \frac{-1}{\sqrt{1-2x}^3}$$

Question 6.

a)



b) (i) $f(x) = \frac{\sqrt{x} + 1}{\sqrt{x} - 1}$

$$f'(x) = \frac{\frac{\sqrt{x}-1}{2\sqrt{x}} - (\frac{\sqrt{x}+1}{2\sqrt{x}})}{(\sqrt{x}-1)^2}$$

$$= \frac{-2}{2\sqrt{x}(\sqrt{x}-1)^2}$$

$$\therefore f'(x) = \frac{-1}{\sqrt{x}(\sqrt{x}-1)^2} \text{ (as required)}$$

(ii) The equation of the tangent has the form: $y - y_1 = m(x - x_1)$ where $(x_1, y_1) = (4, 3)$
 ie $y - 3 = \frac{-1}{2}(x - 4)$
 $2y - 6 = -x + 4$
 $\therefore x + 2y - 10 = 0$ (as required)

c) (i) Weekly rent for n increases
 $= (300 + 20xn)$

No of units occupied
 $= 100 - 4xn$

$$\therefore \text{Rent } (R) = (300 + 20n)(100 - 4n) = 30000 + 800n - 80n^2 \text{ (as required)}$$

(ii) To maximize revenue/rent:

$$\frac{dR}{dn} = 800 - 160n$$

$$\frac{d^2R}{dn^2} = -160 (< 0 \Rightarrow \text{maximum})$$

For turning pts $\frac{dR}{dn} = 0$

$$\text{ie } 800 = 160n$$

$$\therefore n = 5$$

$$R = \frac{30000 + 800 \times 5 - 80 \times 5^2}{100}$$

$$\therefore R = \$320.00$$

So the weekly rent should be \$320 in order to maximize revenue.

d) Find the cartesian equation of the (4) parametric equations

$$x = 1 + 2^{-t} \quad \text{--- (1)}$$

$$y = 2^t \quad \text{--- (2)}$$

From (1) $(2^{-t})^{-1} = (x-1)^{-1}$
 $2^t = \frac{1}{x-1}$

$$\therefore y = \frac{1}{x-1} \neq$$

where $y = 2^t$.

c) Cost/hour = $\$(3 + kV^3)$
 $6.75 = 3 + 3375k$

$$\therefore k = \frac{1}{900}$$

Let x be total running costs:-

$$x = \left(3 + \frac{V^3}{900}\right) \times \frac{450}{V} \quad \text{no. of hours}$$

$$\frac{dx}{dV} = -\frac{1350}{V^2} + V = 0$$

$$V^3 = 1350$$

$$\therefore V = 11.052...$$

Thus, we have a min. cost

when $V < 11$

$$\frac{dx}{dV} < 0$$

when $V > 11$

$$\frac{dx}{dV} > 0$$